

Foldable Robotics: Design, Build, and Simulation Project

Optimization and Experimental Validation of a Grasshopper-
Inspired Jumping Mechanism

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Course: Foldable Robotics

Research Scope & Objectives

The objective of this project is to design, simulate, and experimentally evaluate a bio-inspired foldable robotic leg that is based on a four-bar linkage, mimicking the hind leg of a locust. The study will explore how varying the length ratios between the femur and tibia segments affects jumping performance, focusing on jump height, horizontal distance, and mechanical efficiency.

Research Question: What impact do the length ratios of the femur and tibia segments within a four-bar, locust-inspired jumping mechanisms have on jump height, jump distance, and mechanical efficiency?

Scope: The project involves developing a planar four-bar leg model inspired by the anatomy of a locust's femur and tibia. It includes kinematic and dynamic analysis of the mechanism using simulation tools such as MuJoCo and Python. A parametric study will be conducted, systematically varying the lengths of the femur and tibia (and their ratio) to assess their effects on:

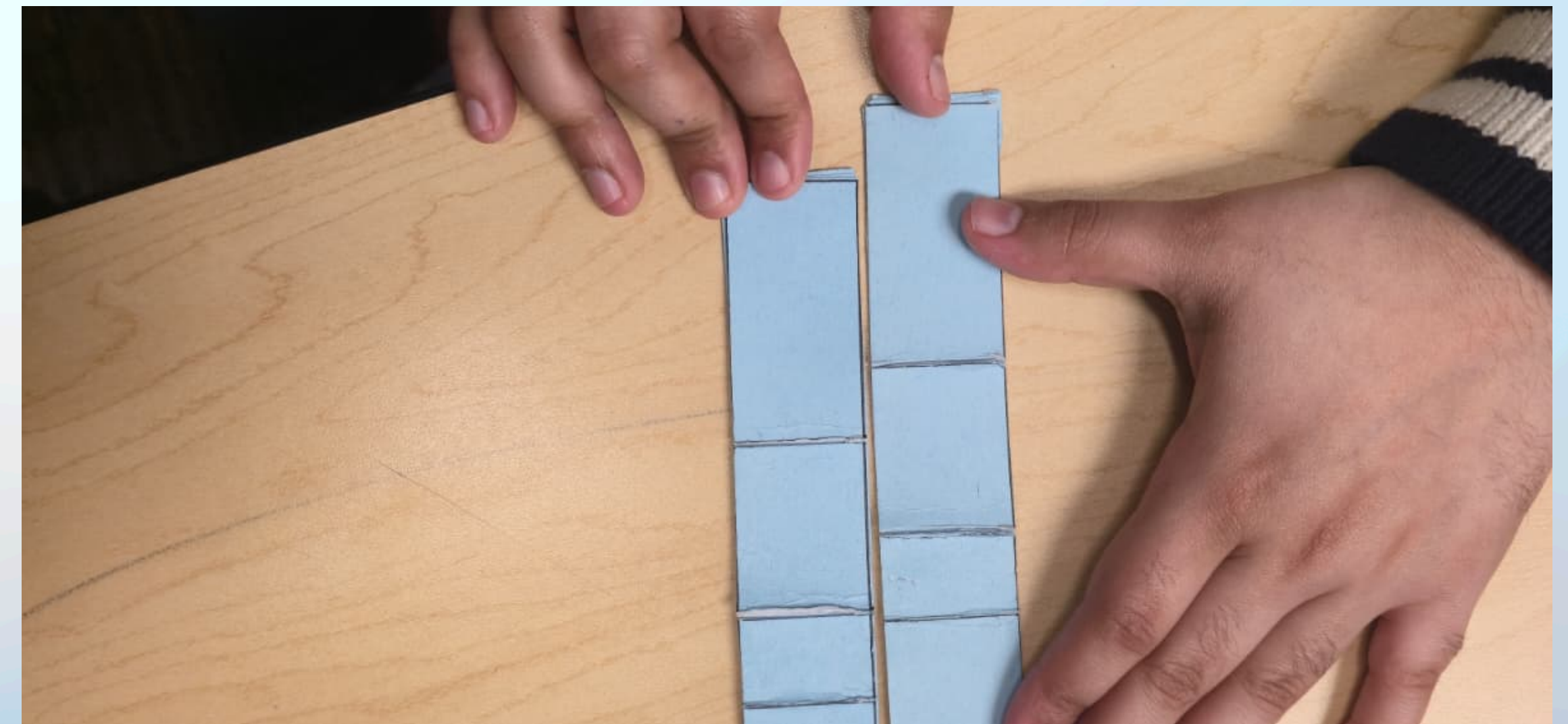
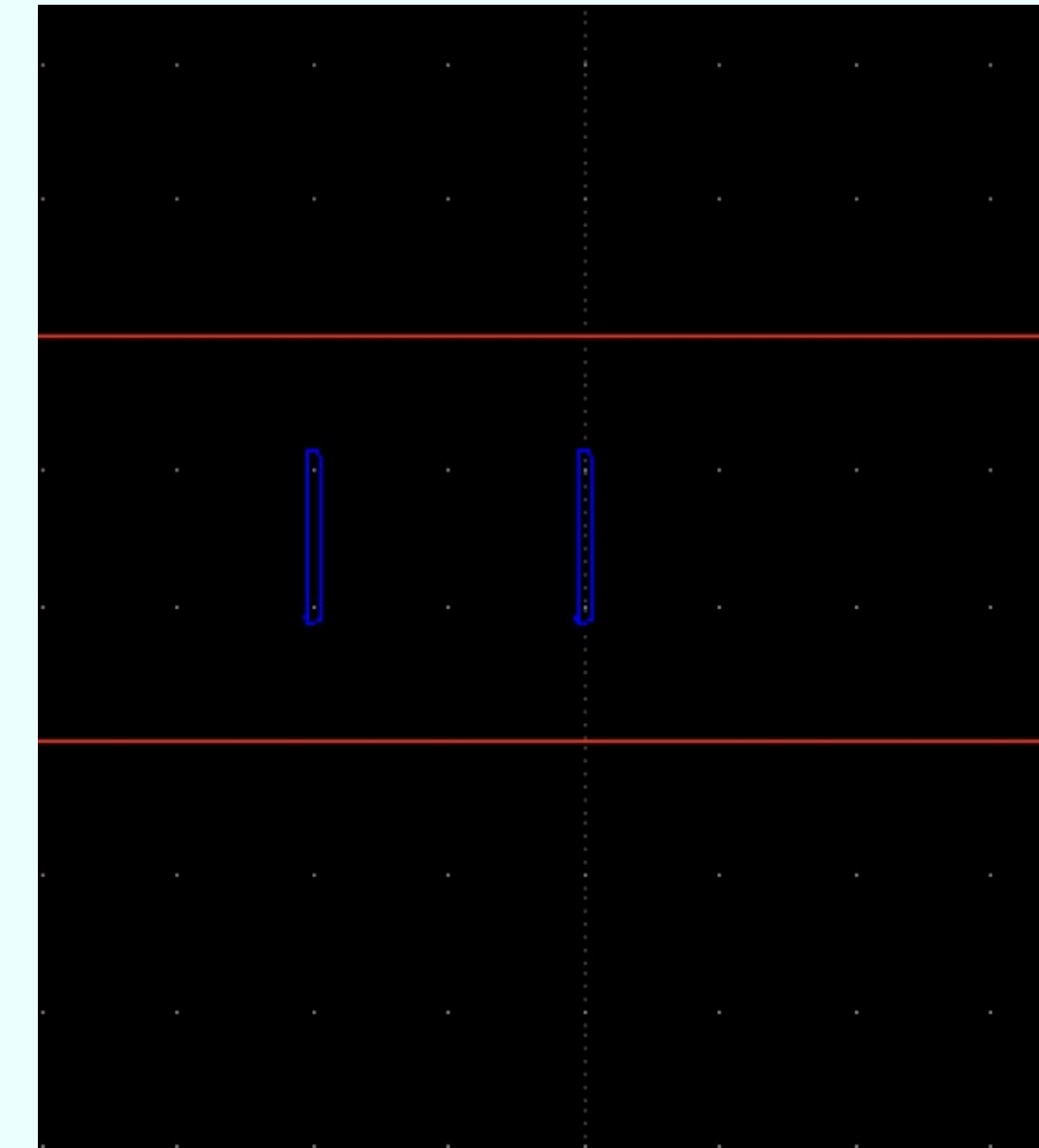
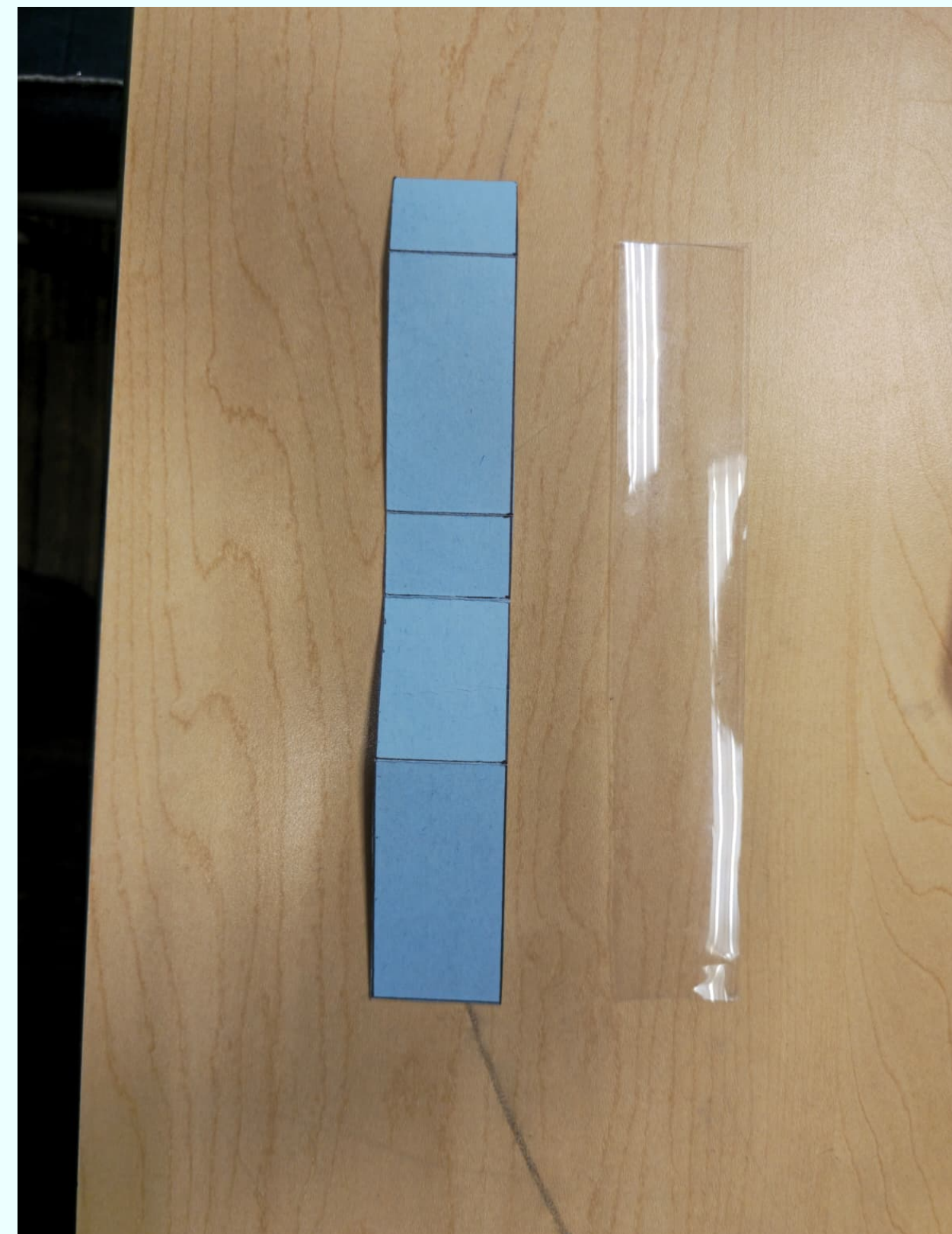
- Maximum jump height
- Horizontal distance
- Energy or work required per jump (mechanical efficiency)

Additionally, the design and fabrication of a foldable physical prototype will be carried out using lightweight materials, including paper, cardboard, acrylic, or 3D-printed components. The prototype's jumping performance will be experimentally tested and compared with the simulation results.

Manufacturing & Fabrication Process

Approach: Layer-based manufacturing using laminate design.

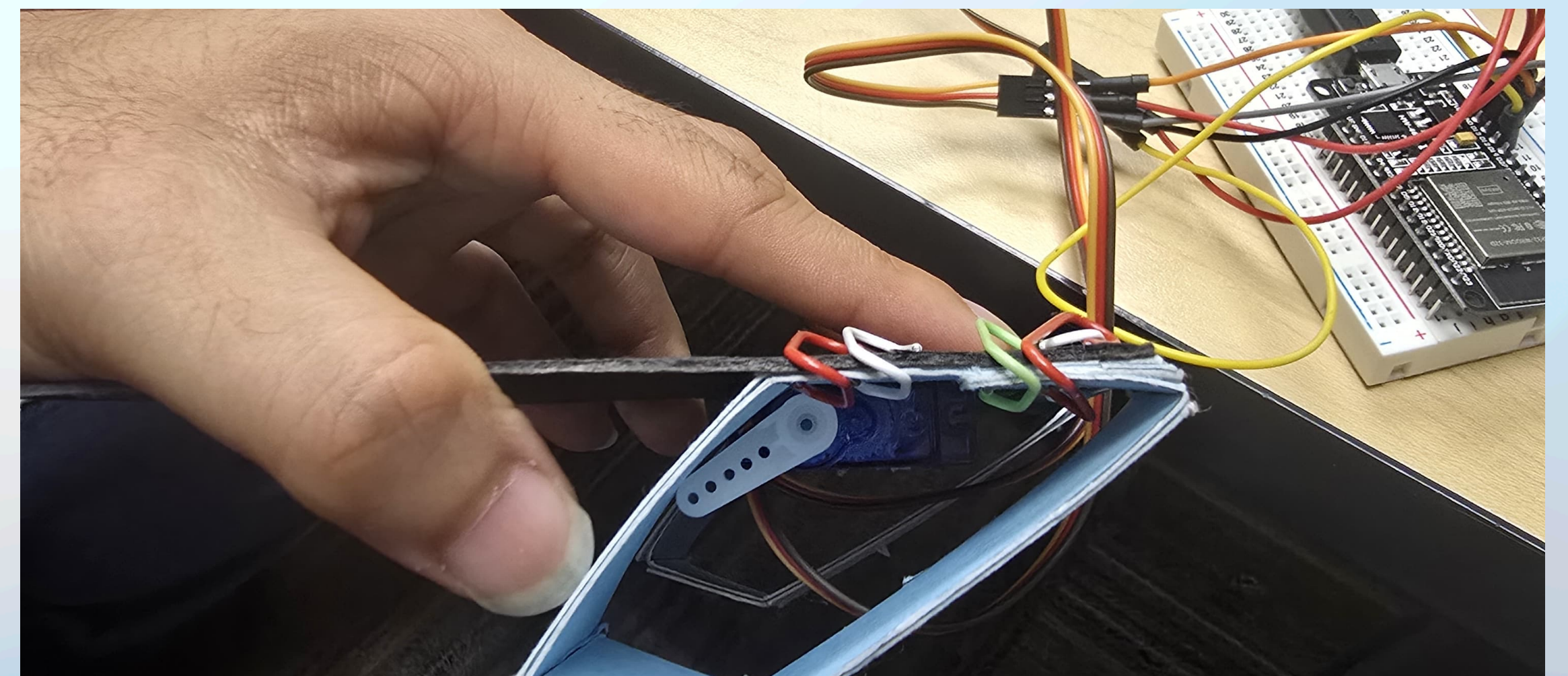
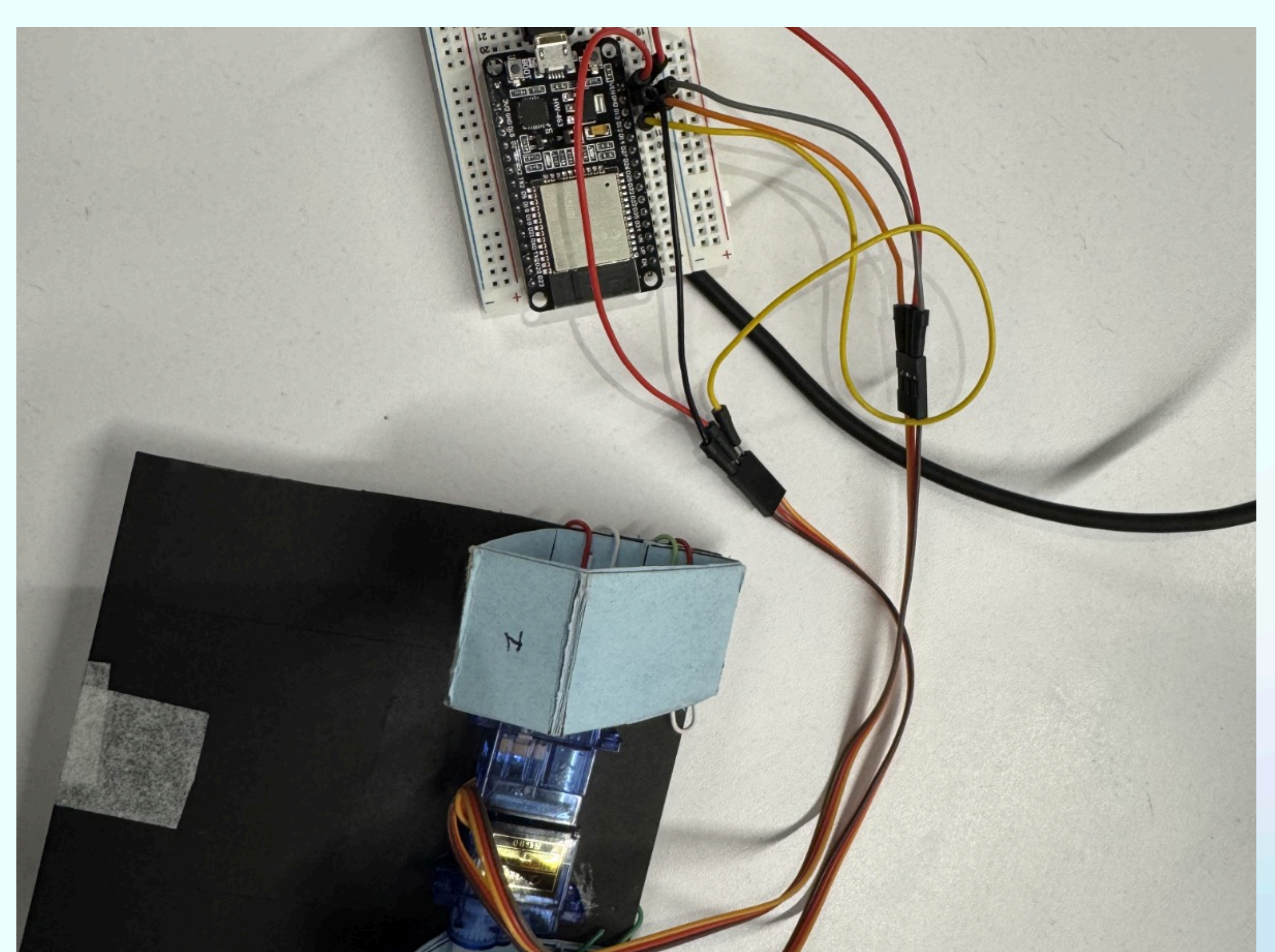
- **Mechanism Design:** Defined geometry using LibreCAD/ DXF with color-coded hinges and cuts.
- **Design for Experimentation:** The prototype was built with adjustable features to allow varying the specific design parameter chosen for optimization.
- **Fabrication Pipeline:**
 - Cutting: Laser cut geometry for each layer.
 - Assembly: laminate layers and fold the system.
 - Electronics: Attached servo motors and integrated ESP32 for gait control.



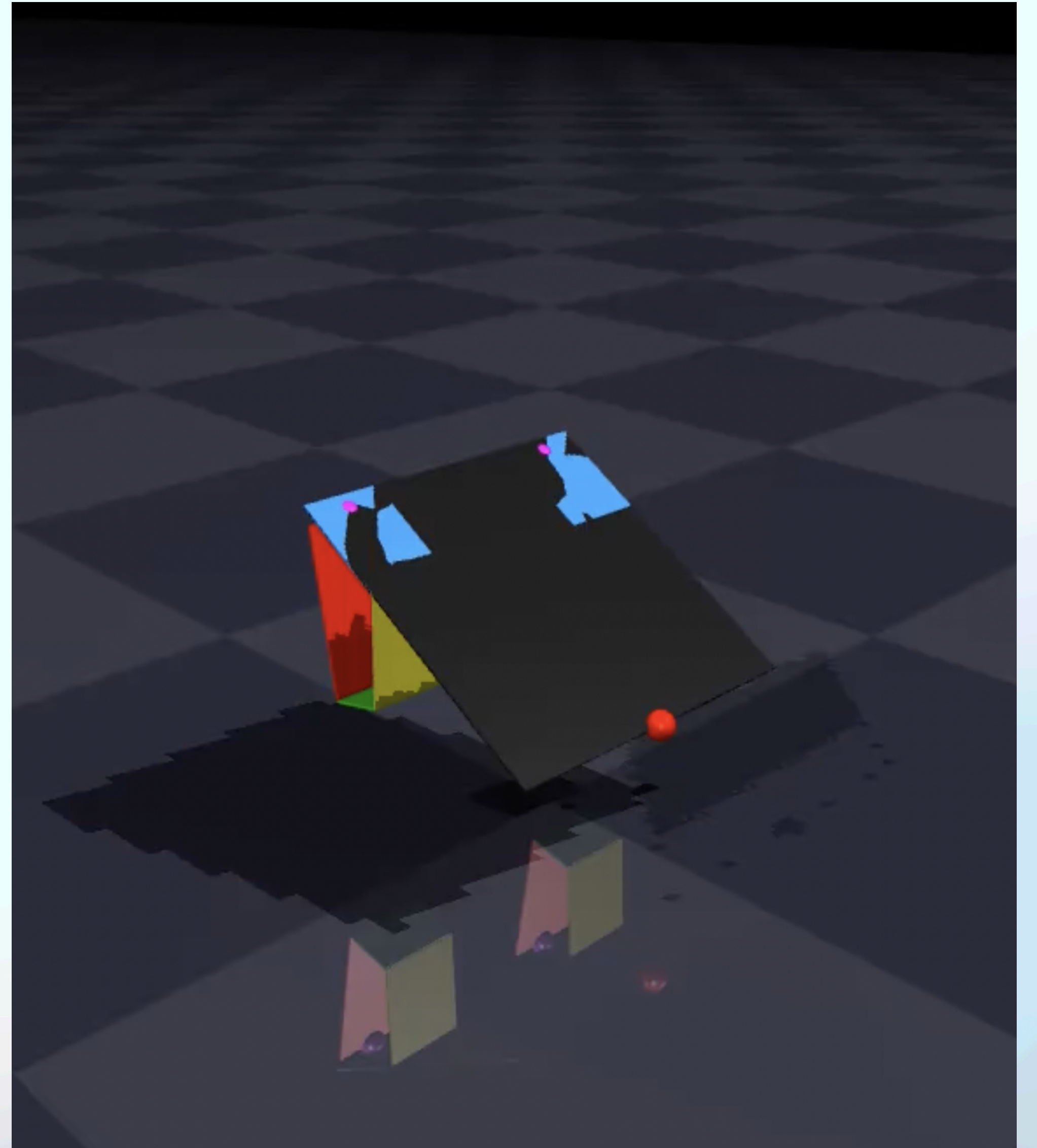
ROBOT MECHANISM

GRASSHOPPER-INSPIRED
MECHANISM

FOUR-BAR MECHANISM
1 DEGREE OF FREEDOM

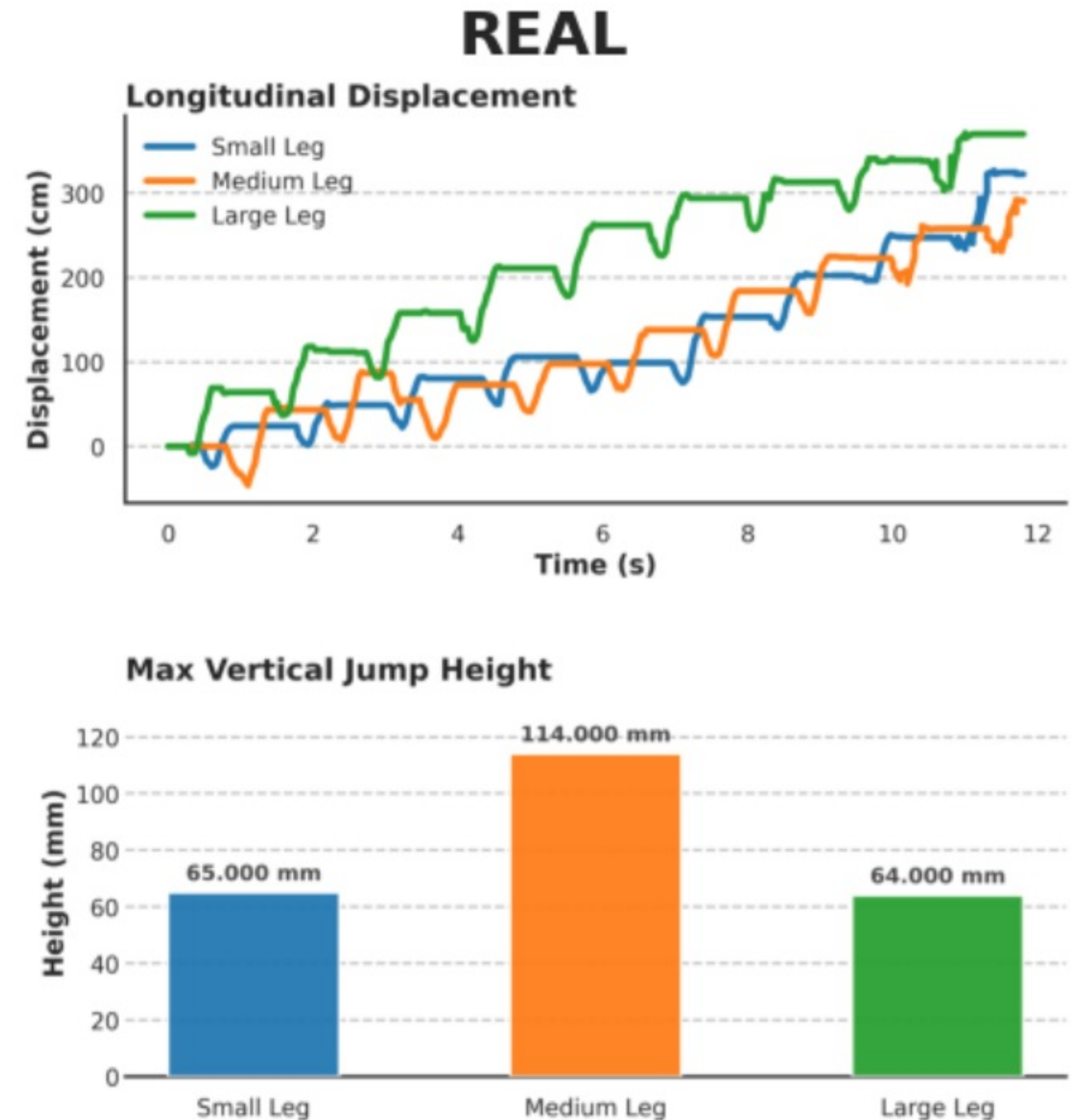
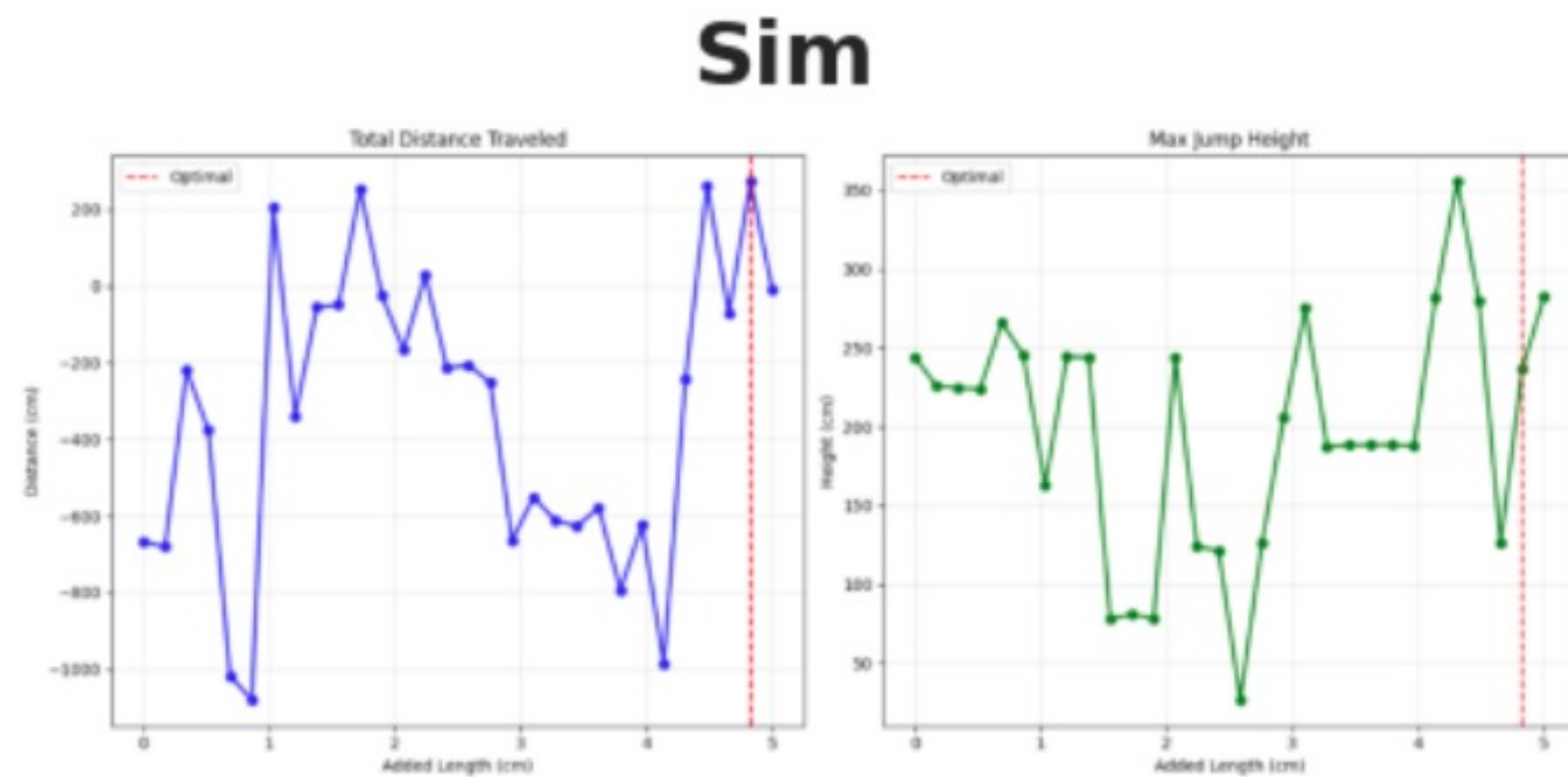


MUJOCO SIMULATION



Experimental Analysis & Validation

Comparing Simulation vs. Reality



OBSERVATIONS AND CONCLUSION

Synthesis of Results The integration of empirical parameter identification with a parametric MuJoCo simulation provided a robust optimization framework for the 4-bar linkage. The "Granular Sweep" revealed a critical design trade-off: stride length vs. actuator torque.

Simulated Optimal: The model predicted a theoretical peak performance at +4.83 cm (Large-scale). At this size, the simulation balanced maximum stride length against the leverage required to lift the chassis.

Physical Optimal: Real-world testing confirmed the trend but highlighted practical limits. The Medium (+1 cm) robot proved superior, offering reliable locomotion. The physical "Large" (+2 cm) robot, while theoretically faster, consistently stalled due to torque limitations, confirming that motor capacity is the primary constraint.

Model Validity & Divergence The simulation accurately predicted mechanical advantage but diverged in two areas: Friction: The constant Coulomb model ($\mu \approx 1.7$) overestimated travel distance by ignoring the variable slip seen during high-impact landings in reality. Compliance: The simulation assumed rigid links with compliant joints. In reality, the "Large" physical legs buckled under load, absorbing kinetic energy and underperforming compared to the rigid-body model.

Final Verdict Geometric scaling effectively enhances mobility, but only within the limits of the actuator. While the simulation points to a theoretical optimum of +4.83 cm, **the Medium Configuration (+1 cm) is the best practical design for the current servo motors.**

Achieving the simulated theoretical performance would require upgrading to higher-torque motors and stiffer composite materials